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Blockchain

Stone Chung, Dean Wu, Isaac Olson, Emi kobayashi, Daiki Matsutaka

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Agenda and Individual Contribution

1. Intro- Isaac (20%)
2. S- Curve- Stone (20%)
3. TOE- Emi (20%) and Daiki (20%)
4. Case Study- Dean (20%)
5. Conclusion- Isaac

Resources

1. Tapscott, Don and Alex. *Blockchain Revolution*. 2016.
2. Nakamoto, Satoshi. “Bitcoin: A Peer-To-Peer Electronic Cash System.” 2008.
3. Casey, Michael J. and Paul Vigna. *The Truth Machine: The Blockchain and the Future of Everything*. 2018.

Intro: What is Blockchain?

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Blockchain is a distributed digital ledger technology that uses a network of users to make data immutable and resistant to digital tampering. There is no central authority for blockchain, it is completely decentralized. Any node on the network maintains a copy of the entire digital ledger, where each block is cryptographically linked to the history of other blocks. The digital structure of blockchain ensures that data cannot be altered without the majority consensus of the network. This technological innovation has disrupted financial markets and financial technology, as well as data storage methods. It's introduction has improved the security and traceability of data transactions while making it exponentially difficult to alter or hack.

Intro: Evolution of Blockchain

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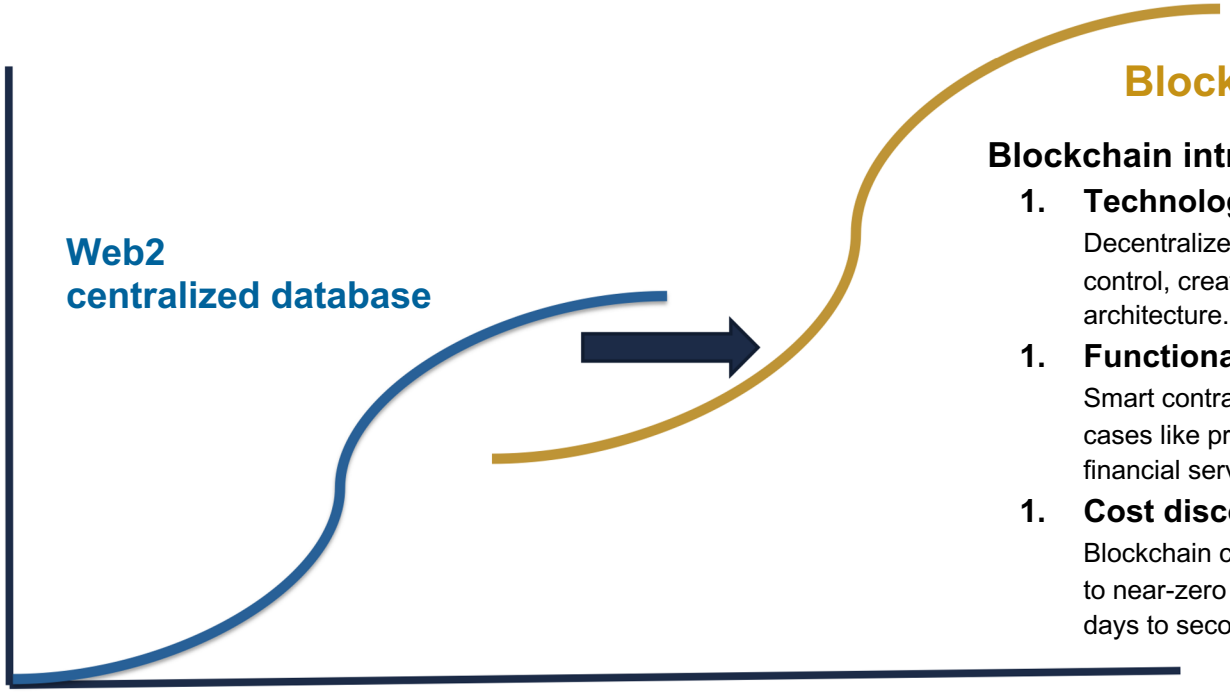
Blockchain Timeline:

1. **1991**
   Stuart Haber and W. Scott Stornetta of Surety Technologies created the first form of blockchain technology which was initially developed as cryptographically secured chain of digital blocks to put timestamps on documents to avoid unauthorized alteration.
1. **1992**
   Harvard mathematician David Bayer Blockchain incorporates Merkle trees, which put multiple documents into one block to improve verification.
1. **2008**
   Bitcoin emerges, combining proof-of-work consensus (among participating nodes) and hash rates to create a completely decentralized ledger, thus creating the first digital currency. This is blockchain as we know it today.

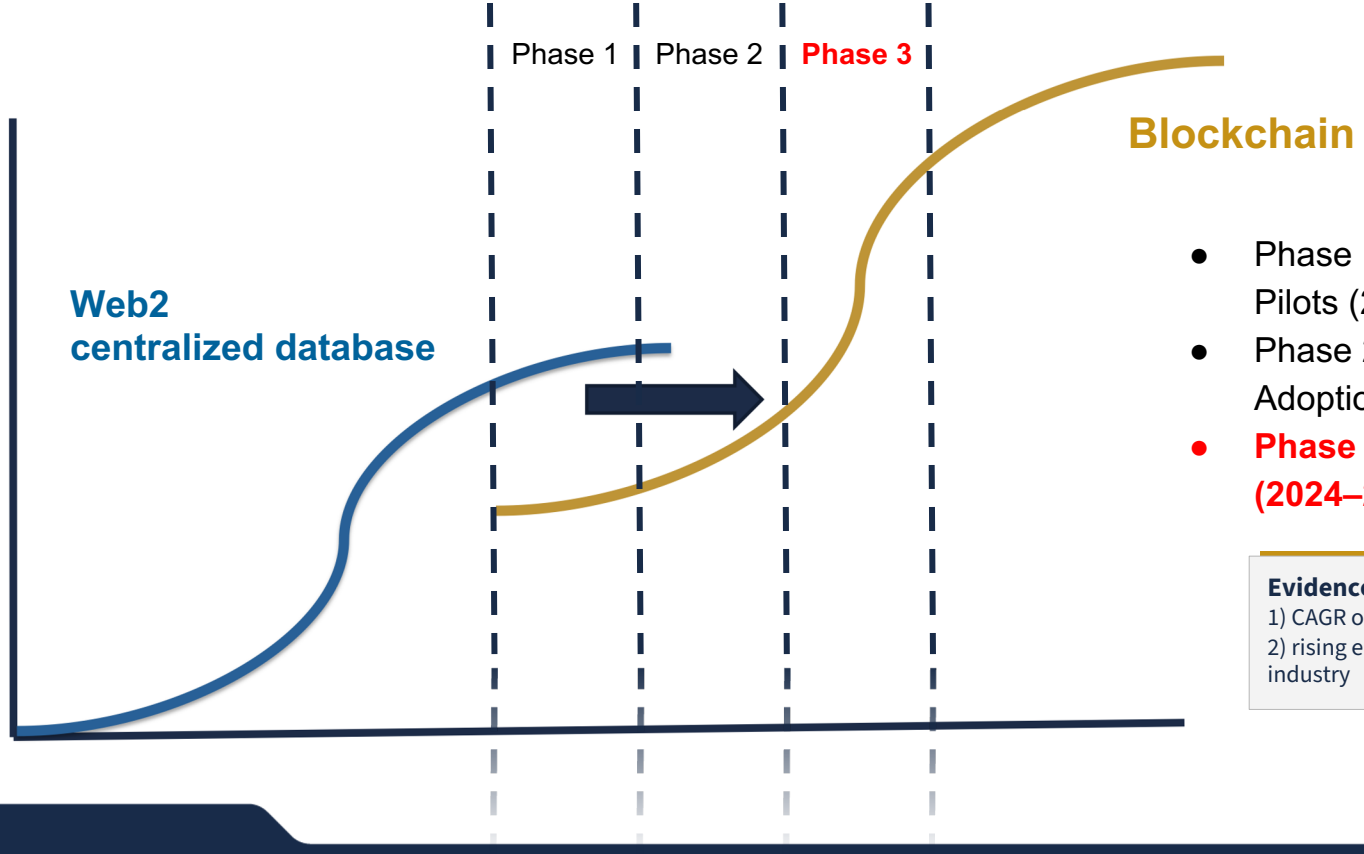
S Curve- Why Blockchain Represents a New S-Curve



Blockchain introduces 3 discontinuities.

1. **Technology discontinuity:**
   Decentralized consensus replaces centralized data control, creating an entirely new infrastructure architecture.
1. **Functional discontinuity:**
   Smart contracts automate trust and enable new use cases like programmable money and composable financial services.
1. **Cost discontinuity:**
   Blockchain cuts transaction costs from multi-percent fees to near-zero and reduces cross-border settlement from days to seconds.

S Curve- Where Is Blockchain on the S-Curve?



- Phase 1： Early Research / Pilots (2015–2020)
- Phase 2： Early Market Adoption (2021–2024)
- **Phase 3： Inflection Zone (2024–2025)**

Evidence
1) CAGR of 90.1% from 2025 to 2030
2) rising enterprise use cases in Muti-industry

S Curve- What Industry Will Be Disrupted?

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Financial Services

Blockchain rewrites the trust and settlement infrastructure of financial services.

- DeFi TVL: Grew from ~$1B (2020) to tens of billions, showing real adoption
- Disintermediation: On-chain payments and settlement bypass clearinghouses

Supply chain and Traceability.

Blockchain replaces siloed databases with a shared, tamper-proof ledger for global trade.

- Global fraud: Counterfeit market > $1.7T; blockchain reduces verification costs
- Real deployments: IBM–Maersk projects automate cross-border documentation

TOE Framework Overview: Blockchain's Transformative Dynamics

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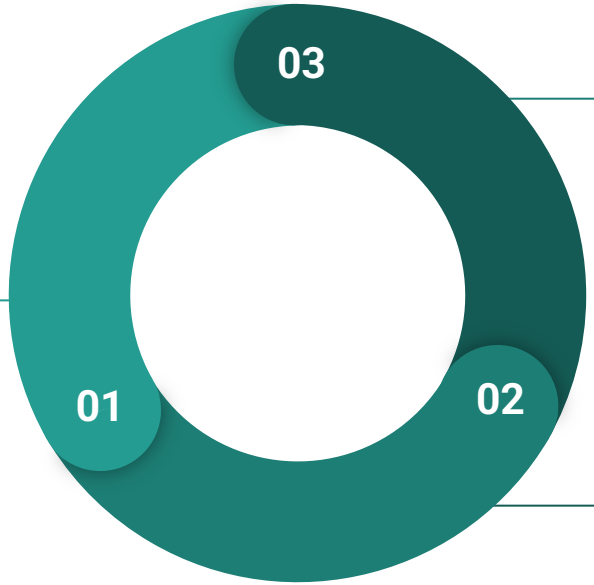


Environmental Dimension (E)

The environmental evolution — regulation, markets, and ecosystems.

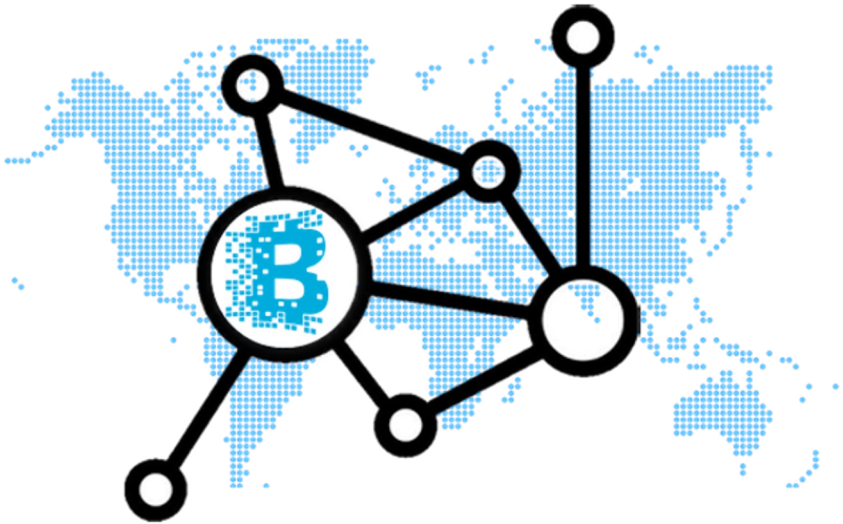
Technological Dimension (T):

The technological foundation enabling decentralized trust. (Blockchain Infrastructure)

Organizational Dimension (O)

The organizational adaptation toward hybrid finance. (DAOs, Fintech, Banks)

Technological Dimension (T)



- **Decentralized Ledger:** transactions verified by network consensus, not a central bank.
- **Proof-of-Work Consensus:** cryptographic validation builds trust through code.
- **Transparency & Immutability:** once recorded, data can't be altered.
- **Lightning Network:** second-layer innovation improving speed and cost.
- **Challenges:**
  - ~7 TPS vs Visa's 24,000 TPS
  - High energy use (~90 TWh/year)
- **Ongoing Innovations:** Proof-of-Stake, Layer-2 scaling → more efficient, scalable, secure.
- **Impact:** Blockchain redefines financial infrastructure — trust without intermediaries.

Organizational Dimension (O)

O1: Traditional Finance

- Centralized banking, slow innovation.
- Hierarchical decision-making.
- Resistant to decentralization.

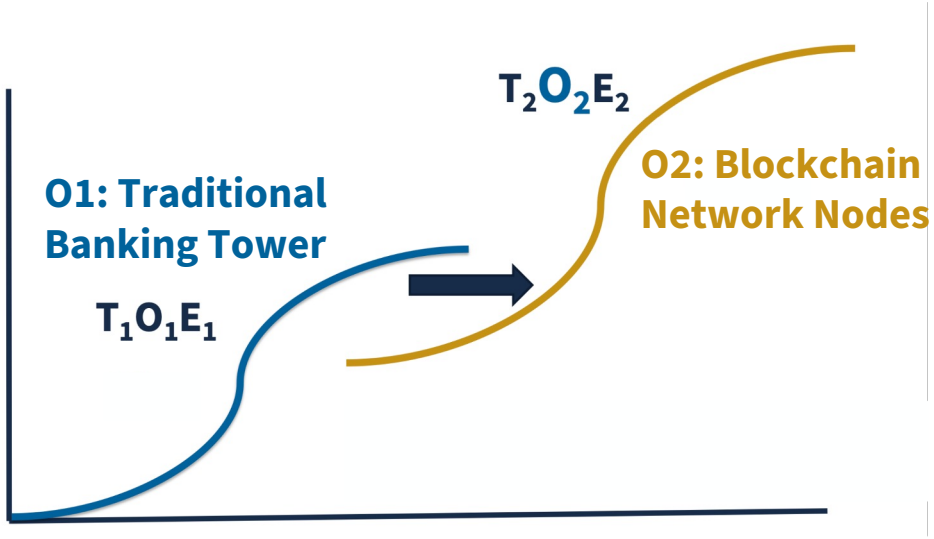
O2: Emerging Models

- DAOs (Decentralized Autonomous Organizations).
- Corporate adoption: BlackRock, Fidelity, Tesla, MicroStrategy.
- Fintech integration: Coinbase, PayPal, Revolut.
- Central Bank Digital Currencies (CBDCs).

Shift: From control → collaboration → hybrid financial systems.

Cultural Change:

- Traditional finance embracing risk and experimentation.
- Collaboration with fintechs for custody, compliance, infrastructure.

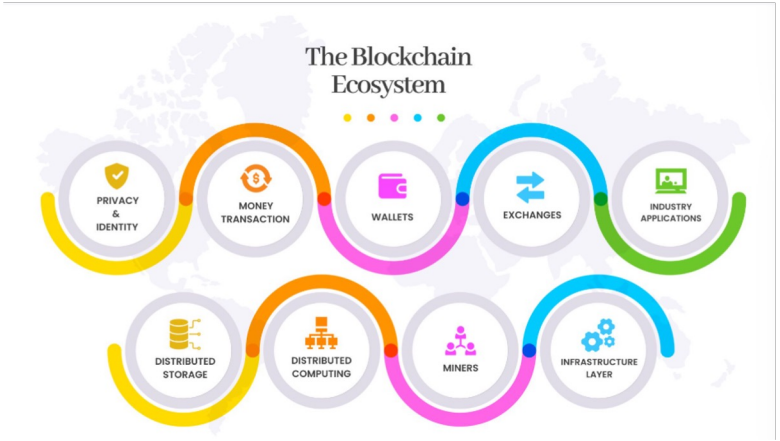


Environmental Dimension (E)

We use Bitcoin as the representative example to illustrate environmental barriers to blockchain adoption.

Regulation

- Global rules remain fragmented
  - **U.S.:** SEC uncertainty delays institutional adoption
  - **EU:** MiCA provides clear licensing → accelerates adoption
  - **China:** Full ban limits global network effects
- Regulatory inconsistency slows enterprise deployment



Market Competition & Industry Structure

- Strong incumbents (Visa, Mastercard, SWIFT) maintain network effects
- Competing technologies: Ethereum/L2s, Solana, CBDCs, private chains
- Institutional entry (BlackRock, Fidelity, PayPal) boosts legitimacy
- Competition accelerates innovation but increases fragmentation

Social & Institutional Acceptance

- ESG criticism toward PoW energy consumption
- Public trust issues: volatility, hacks, scams
- Enterprises cautious due to compliance and reputation risk
- Younger demographics increasingly crypto-native → long-term tailwind

TOE Alignment Summary & Investment View

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TOE Alignment Score

- **Technology (T): 8/10**
  Decentralized, secure, and increasingly scalable (L2, PoS innovations)
- **Organization (O): 6/10**
  Incumbents experimenting but risk-averse; emerging models (DAOs, fintechs) accelerating
- **Environment (E): 6/10**
  Regulation remains the dominant constraint; societal acceptance uneven

Overall TOE Alignment: 7/10

Bitcoin's technological foundation is strong, and institutional adoption is rising, but **regulation and trust barriers remain the primary constraints** on mass adoption.

Investment View: Selective YES (Long-Term)

- Strong technological trajectory
- Institutional adoption accelerating
- Regulatory progress slow but improving
- Long-term S-curve potential as foundational financial infrastructure
- Short-term volatility and environmental misalignment remain risks

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Goldman Sachs

Bitcoin & Blockchain vs. Goldman Sachs AI Assistant

Disruptive digital infrastructure in finance	Decentralized ledger	Firm-wide GenAI platform
Technological Context	Advanced infrastructure & Tech readiness	
Organizational Context	Develop new internal capabilities and governance systems	
Environmental Context	Regulation & Reputation Risk	

What Blockchain Can Learn from GS AI Assistant Assimilation

How does a new digital technology reshape perceptions of money and trust?

Trust Mechanisms

Centralized vs. decentralized innovation paths

Assimilation Phases

Initiation

Adoption

Routinization

Bitcoin Blockchain vs. NVIDIA CUDA

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Industry impact

Both are foundational technology platforms

Ecosystem & network effects

Technological benefits

What CUDA's Diffusion Teaches Us About Blockchain Adoption

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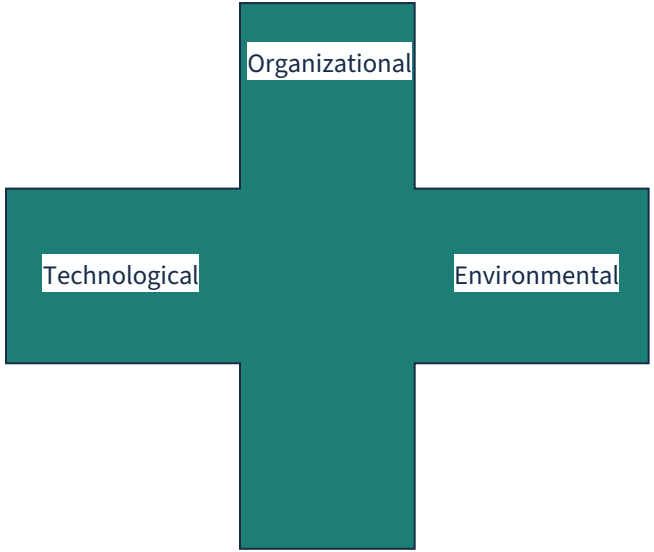
Managerial Obstacles

High Performance + Mature Tooling



Regulatory Uncertainty

Conclusion

Blockchain technology has revolutionized how data is managed and verified over the past 10 years. It has disrupted financial markets, but is also being adopted in healthcare, supply chains, and various other data managing services. It's proprietary method of proof-of-work consensus ensures data is immutable and transparent on a decentralized ledger and does not require a central authority to operate it. It will face challenges of acceptance and regulation from various governments, but could become a foundation for digital transactions across the globe on a new S-Curve.

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Thank you!